ARGUS NEO: Trust-Gated Counterfactual Dual Control for Adaptive Acoustic Structural Interrogation

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***Abstract*—ARGUS NEO is an inspectable research prototype for sequential acoustic interrogation of a panel with one dominant scatterer. It maintains a spatial posterior and separate metrology uncertainty, including correlated velocity and timing estimates; selects finite source–receiver–waveform candidates; and supports diagnosis, direct-reference calibration, verification, and abstention. The default controller uses labelled heuristic utilities. An optional controller compares calibration and diagnosis using expected squared localization error plus acquisition cost in square metres, with Gaussian observation branches over at most two measurements. Quality, accumulated channel reliability, and out-of-distribution checks precede structural fusion. Evidence exports support integrity verification against an independently retained checkpoint, not measurement authentication. In a historical September synthetic check of six paired seeds per condition and six measurements per case, nominal mean errors were 43.90 mm for the legacy controller and 12.58 mm for the optional controller; under velocity, attenuation, and noise mismatch they were 150.22 and 203.78 mm. Both paired bootstrap advantage intervals included zero. These results do not validate the changed October source. The optional controller remains off by default. The October revision adds bounded acquisition/import contracts, network controls, and scientific regression cases. The implementation does not establish physical accuracy, a general joint posterior, commercial superiority, or legal patentability.**

***Index Terms*—active sensing, Bayesian experimental design, dual control, structural health monitoring, nondestructive evaluation, uncertainty quantification, digital twin, out-of-distribution detection, adaptive measurement, guided waves**

I. Introduction

Nondestructive evaluation (NDE) is usually presented as a mapping from a measurement to a decision: acquire a waveform, compute features, and classify or localize damage. That formulation leaves the measurement geometry fixed. Yet the source position, receiver position, waveform, frequency band, amplitude, and duration determine which aspects of an internal structure are observable. When measurements have different costs, choosing the next experiment can be as important as interpreting the previous one.

ARGUS addresses the sequential question

$$e_{t+1}^{\star} = \arg\max_{e \in \mathcal{E}_t} \left[\mathcal{I}_t(e) - \mathcal{C}_t(e) \right], \tag{1}$$

where e is a physically executable experiment and $\mathcal{E}_t$ is a state-dependent candidate set. After observing response y_{t+1}, ARGUS performs a recursive Bayesian update, regenerates candidates from the new posterior, and repeats. This is related to Bayesian experimental design and active data selection [1], [2], but instantiated as a live, spatial, source–receiver interrogation loop.

TABLE I
Passive Detection Versus ARGUS Active Interrogation

Property	Passive pipeline	ARGUS prototype
Measurement	Fixed plan	Chosen from current belief
State	Label or point	Normalized spatial posterior
Uncertainty	Often implicit	Entropy, covariance, local mass
History	Optional aggregation	Recursive Bayesian fusion
Output	Detect or locate	Map, next action, reason, stop state
Efficiency	Scan or fixed network	Information–cost tradeoff

The prototype targets an opaque rectangular panel with one dominant hidden anomaly (cavity, loose region, delamination, or dense inclusion). Its central distinction from a one-shot classifier is summarized in Table I. It is designed to run without hardware through a seeded physics-inspired simulator. Physical recordings are accepted for history and analysis, but arrival inference additionally requires supported excitation-relative timing and a declared validated arrival adapter.

The contributions of this implementation are:

1) an observe–infer–plan–act loop joining simulation or acquisition to block-structured spatial and metrology estimates, retaining velocity–timing covariance rather than claiming a general joint posterior;
2) a dual-control policy that switches among diagnostic, calibration, verification, and abstention actions and evaluates counterfactual observation distributions using risk, worst-rival separation, information, movement, energy, redundancy, trust, and safety terms;
3) selection among waveform families and source–receiver geometries, with two live analytical/physics-signature fidelity levels; the legacy route lookahead is a geometric proxy, while the optional risk controller branches on observations;
4) likelihood tempering, online model-discrepancy correction, OOD indicators, calibration diagnostics, and

explicit verify/abstain states, without claiming field-calibrated probabilities;
5) an auditable research stack comprising resumable jobs, paired benchmarks, ablations, fault injection, truth-sealed replay, hash-chained evidence export, phone-camera guidance, and a reconnecting edge node;
6) tested endpoints, numerical and safety regression tests, and a production frontend build; the dated audit report records the current verification evidence.

A. Revision Scope: 7 October 2026

The following architectural equations describe modelling intent unless explicitly identified as executed. The runtime is block-structured, not a fully coupled structural–metrology posterior. Learned-surrogate and imported-response-bank fidelity levels are not validated live-controller implementations. Multimodal sensing, robotics, multi-defect localization, and reliable defect type, depth, or severity inference are outside the demonstrated scope. The terms "dual control" and "Bayes risk" in the legacy formulation denote engineered proxies; the distinct, optional physical-loss controller below is the implemented finite-horizon Bayesian design approximation. Healthy/damage/unknown scores and trust values are also heuristic, not certified probabilities.

A direct-path observation satisfies $y = d/v + b + \epsilon$, with propagation state $\theta = (v, b)$ and covariance C. Its Jacobian is $h = (-d/v^2, 1)$. The Gaussian covariance update is implemented in Joseph form,

$$C^+ = (I - Kh)C(I - Kh)^T + KRK^T, \tag{2}$$

$$K = Ch^T(hCh^T + R)^{-1}. \tag{3}$$

A single delay supplies rank-one information; different reference path lengths are necessary to separate velocity from timing. Absolute pose requires independently measured pose errors and uncertainty. Waveform quality alone is not treated as separate evidence for both transmitter and receiver coupling.

For spatial regions with probabilities p_j, centres x_j in metres and unresolved within-region variances s_j^2, the optional controller uses

$$r(p) = \sum_j p_j(\|x_j\|^2 + s_j^2) - \left\| \sum_j p_j x_j \right\|^2. \tag{4}$$

All posterior mass contributes. Calibration and diagnosis share this loss and a declared cost conversion. Within a finite beam of candidate actions, the two-step objective is

$$\begin{aligned} Q(a) = c(a) + \mathbb{E}_{y|a} \min \big\{ & r(p_y), \\ & \min_b [c(b \mid a) + \mathbb{E}_{z|y,a,b} r(p_{y,z})] \big\}. \end{aligned} \tag{5}$$

Conditioning the second observation includes the shared velocity–timing cross-covariance. The second-step options are diagnostics or stopping; this is not unrestricted multi-step optimal control. Independent measurement and pose noise, linearized propagation, region coarsening, and conditional within-region position independence remain assumptions. Gauss–Hermite orders 16 and 32 provide a reported numerical sensitivity diagnostic, not a rigorous integration error bound. Planning's arrival model and inference's generalized matched-filter likelihood remain different approximations.

OOD assessment now precedes fusion. Rejected or stale verification labels cannot satisfy the stopping gate. A successful direct reference validates neither the scattered-wave model nor field coverage, and fitted calibration residuals no longer populate the structural conformal reference pool. Integrity verification checks stored acquisition content; a separately retained ledger head and count are needed to detect whole-history replacement. Checksums alone do not establish acquisition authenticity.

The candidate technical contribution is deliberately narrow. Guided-wave NDE, Bayesian localization, Bayesian sensor placement, active learning, dual control, matched filtering, and active digital-twin calibration all predate this work [3]–[7]. The implementation proposed for examination couples live rival-mode discrimination, structural–metrology action arbitration, constrained candidate selection, gated evidence, and verifiable decision history. Neither technical novelty nor legal patentability is established by this manuscript.

II. Related Work and Design Position

A. Guided-Wave and Vibration-Based Inspection

Elastic waves can interrogate plate-like structures over areas larger than a probe footprint, while damage modifies propagation through scattering, attenuation, mode conversion, and local resonance [3]. Practical guided-wave interpretation is complicated by dispersion, anisotropy, boundaries, coupling variability, and environmental drift. Open Guided Waves provides controlled CFRP measurements for comparative algorithm research [8], illustrating both the value and the complexity of measured wavefields.

ARGUS does not claim that its compact simulator is a high-fidelity Lamb-wave or finite-element model. It uses path length, propagation delay, attenuation, signed defect reflection, frequency-sensitive gain, and damped ringing to create an interpretable inverse problem that can exercise the complete adaptive loop. The real-device path is intentionally decoupled from this simulator.

B. Bayesian Design and Active Measurement

Bayesian experimental design selects an experiment according to expected utility under current uncertainty [1]; information-based active selection chooses data expected to be informative about hypotheses or predictions [2]. In structural health monitoring, Flynn and Todd formulated optimal actuator/sensor placement using Bayes risk and demonstrated active guided-wave examples [4]. Much of that literature optimizes a fixed network or configuration. ARGUS instead replans a single next, executable experiment after every observation and incorporates movement, excitation, and redundancy costs.

Exact expected information gain marginalizes future observations for each candidate and depends on the generative model. The default fast path uses a bounded counterfactual-disagreement proxy under its legacy "expected information gain" name. The optional nuisance-arrival extension instead numerically integrates information for a declared regional scalar model; neither calculation guarantees global optimality or information gain under real wavefields.

C. Open Experimental Evidence

The LMSD 2021 dataset contains measured frequency-response functions (FRFs) from a $600 \times 600 \times 4$ mm cross-ply CFRP plate, a 12×12 coordinate grid, seven accelerometer/hammer nodes, a healthy baseline, and six added-mass scenarios [9]. It supplies real multistatic responses and known mass locations, but not cavities, delaminations, or adaptively selected trials. ARGUS uses it to test data ingestion and feature transfer, not to imply physical validation of every supported defect type.

D. Crowded Prior Art and Commercial Position

The field is crowded at both the apparatus and inference levels. Embedded Lamb-wave arrays, phased focusing, baseline comparison, multi-path prediction, Bayesian damage localization, and Bayesian actuator/sensor placement are established [4], [5], [10]–[12]. Autonomous metrology programs also combine physics-informed learning, active experiment design, and adaptive calibration [13]; EP 4 592 907 A1 describes recursive Bayesian digital-twin calibration and active instrument-setting adjustments [7]. Accordingly, neither "AI with ultrasound," "Bayesian guided waves," "choose another measurement," nor "self-calibration" is asserted as novel.

Table II compares public product descriptions, not confidential implementations. A blank or negative distinction means only that the reviewed public material does not establish that feature. Commercial systems have field maturity, hardware qualification, trained operators, and application evidence that ARGUS presently lacks.

III. Problem Formulation

Let the panel occupy normalized coordinates $\Omega = [0,1]^2$. Physical distance is evaluated after the transformation

$$\boldsymbol{p}_{\mathrm{m}} = \mathrm{diag}(W, H)\boldsymbol{p}, \tag{6}$$

where the default width and height are $W = 0.60$ m and $H = 0.40$ m. The latent defect state is

$$z = (x, y, r_x, r_y, q, T), \tag{7}$$

containing center, elliptical radii, severity $q \in [0,1]$, and type T. The current prototype updates only location (x, y) on a $G \times G$ grid, conditional on one dominant scatterer. Simulator randomization varies the other attributes, but the displayed size, severity, and type values remain unchanged priors; there is no measured attribute likelihood or defect-existence model.

An experiment is

$$e = (\boldsymbol{s}, \boldsymbol{r}, f_0, f_1, a, d, w), \tag{8}$$

where $\boldsymbol{s}$ and $\boldsymbol{r}$ are normalized source and receiver coordinates, f_0, f_1 are band endpoints, a is normalized amplitude, d is duration, and w is a member of the finite waveform library. The historical baseline used impulse, sine, and chirp families. Given observation history $\mathcal{D}_t = \{(e_i, y_i)\}_{i=1}^{t}$, the belief grid is

$$P_t(j) = P(z_{xy} = j \mid \mathcal{D}_t), \qquad \sum_{j=1}^{G^2} P_t(j) = 1. \tag{9}$$

The initial belief is uniform. Each observation supplies a likelihood $L_{t+1}(j)$ and

$$P_{t+1}(j) = \frac{P_t(j)\max[L_{t+1}(j), \varepsilon]^{\beta_t}}{\sum_k P_t(k)\max[L_{t+1}(k), \varepsilon]^{\beta_t}}, \quad \varepsilon = 10^{-12}. \tag{10}$$

Here $\beta_t \in [0,1]$ is the executed evidence gate, with zero for rejected, unreliable, OOD-blocked, or calibration-only structural evidence. Probability normalization uses scaled/log weights and preserves supported zero/nonzero distinctions. The normalized entropy is

$$\bar{H}(P) = -\frac{\sum_{j:P(j)>0} P(j)\log_2(P(j))}{\log_2(G^2)} \in [0,1]. \tag{11}$$

IV. System Architecture

Figure 1 shows the implemented data flow. Ground truth exists only inside simulation sessions and remains server-side until an explicit reveal action. Both simulated and physical signals pass through the same analysis, persistence, and user-interface contracts.

The Python backend contains domain models, simulation, signal analysis, belief inference, planning, evaluation, device adapters, session orchestration, and SQLite persistence. A FastAPI layer exposes session, recommendation, experiment, upload, posterior, history, reveal, device, and benchmark endpoints. The TypeScript/React frontend visualizes the posterior heatmap, recommended geometry, signal, spectrum, power spectral density, spectrogram, belief history, benchmark curves, and camera overlay. A single launcher starts both processes locally.

V. Physics-Inspired Forward Model

A. Excitation

For $N = \lfloor df_s \rceil$ samples at default sample rate $f_s = 16$ kHz, the excitation occupies the first

$$N_b = \lfloor f_s \min(10\,\mathrm{ms}, 0.22d) \rceil \tag{12}$$

samples. A Tukey window with shape parameter 0.45 suppresses burst discontinuities [19]. A sine uses $\sin(2\pi f_0 t)$; a chirp sweeps linearly from f_0 to f_1; and the impulse option uses a Ricker-like pulse

$$u(t) = a[1 - 2\phi(t)^2]\exp[-\phi(t)^2], \tag{13}$$

$$\phi(t) = \frac{\pi(t - 0.28d_b)}{\max(f_c^{-1}, 0.22\,\mathrm{ms})}. \tag{14}$$

TABLE II
PUBLICLY DESCRIBED MARKET POSITION VERSUS THE ARGUS RESEARCH WEDGE

Platform	Acquisition pattern	Publicly described strength	TG-CDI distinction from reviewed material	ARGUS disadvantage
Gecko Cantilever [14]	Dense robotic ultrasonic/camera acquisition	Large-scale high-fidelity asset data, digital representation, analytics, repair planning	Scores finite next-interrogation candidates from the live posterior in a sparse-acquisition research workflow	No industrial robot fleet, deployment history, or proven inspection throughput
Evident OmniScan X3 [15]	Phased-array FMC/TFM/PCI with electronic focusing	High-resolution imaging, large apertures, fast TFM, qualified field instrument	Selects movable bistatic/waveform candidates with separate location/metrology state, calibration, and stop rules	Far lower acoustic fidelity, channel count, resolution, and certification maturity
Eddyfi Sonyks/Teletest [16]	Long-range guided-wave pipe screening from a deployed ring/tool	Rapid long-distance corrosion screening with established procedure and training	Posterior-conditioned sequential experiment design targets unresolved competing hypotheses on a bounded surface	Does not match pipe-specific range, tooling, standards, or field credibility
Acellent SMART Layer [17]	Permanently bonded distributed PZT network; active or passive operation	Repeatable sensor geometry and integrated lifecycle SHM	Treats source/receiver placement and excitation as reconfigurable decision variables and explicitly arbitrates calibration versus diagnosis	Lacks permanent-network repeatability, ruggedization, and long-duration evidence
MISTRAS AEScout [18]	Passive acoustic-emission monitoring, up to four localized channels	Detects and locates active damage mechanisms during operation	Actively creates the next discriminating wave experiment and can request verification when passive evidence is unsupported	Cannot replace continuous AE monitoring or expert interpretation of active cracking
ARGUS NEO	Sparse movable source/receiver; sequential active excitation	Inspectable TG-CDI policy, low-channel reconfigurability, uncertainty, trust gating, audit trail	Candidate research/licensing layer for measurement-budget-limited or access-constrained inspection	Research prototype; no probability-of-detection study, blind-panel validation, certification, or production hardware

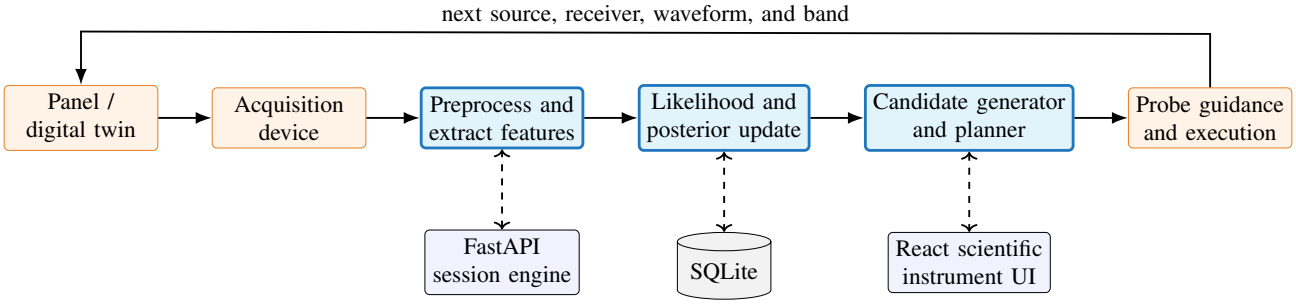


Fig. 1. ARGUS closed-loop architecture. The planner changes the next physical experiment; it does not merely classify a fixed waveform.

B. Direct and Scattered Responses

Let d_{sr} denote physical source–receiver distance and let d_{sz}, d_{zr} denote the two legs through the defect. The response is

$$y(t) = A_0 u(t-\tau_0) + A_z u(t-\tau_z) + r_z(t) + n(t) + b(t), \quad (15)$$

with fractional delays obtained by linear interpolation. For wave velocity c, attenuation α, and system delay t_0,

$$\tau_0 = t_0 + d_{sr}/c, \qquad A_0 = \frac{e^{-\alpha d_{sr}}}{1+4d_{sr}}, \quad (16)$$

$$\tau_z = t_0 + (d_{sz} + d_{zr})/c. \quad (17)$$

The defect resonance and scattered gain are

$$f_z = f_{\mathrm{mat}}(1.05 - 0.32q), \quad (18)$$

$$R_f = 0.72 + 0.72\exp\left[-\frac{1}{2}\left(\frac{f_c - f_z}{1250}\right)^2\right], \quad (19)$$

$$A_z = 0.43\rho_T q \frac{\sqrt{r_x r_y}}{0.085}\frac{e^{-\alpha(d_{sz}+d_{zr})}}{1+5(d_{sz}+d_{zr})}R_f, \quad (20)$$

where $\rho_T = 1.00, 0.72, 0.86, -0.64$ for cavity, loose region, delamination, and dense inclusion, respectively. The negative dense-inclusion coefficient creates a phase inversion. Starting at τ_z,

$$r_z(t) = 0.11 A_z a \sin[2\pi f_r(t-\tau_z)]e^{-\beta(t-\tau_z)}, \quad (21)$$

where $f_r = f_z$ except for dense inclusions, for which $f_r = 1.12 f_z$. Noise $n(t)$ is Gaussian; $b(t)$ is a cumulative low-frequency random drift. The default material values are $c = 180$ m/s, $\alpha = 1.65$ m^{-1}, $f_{\mathrm{mat}} = 3100$ Hz, $\beta = 95$ s^{-1}, $t_0 = 0.8$ ms, and noise standard deviation 0.007.

This model creates a fast, interpretable inverse problem for candidate evaluation. It omits dispersion branches, anisotropic stiffness, boundary-mode conversion, transducer electromechanics, and spatially varying coupling. Geometry can remain ambiguous, and simulation accuracy must not be confused with field accuracy.

VI. SIGNAL INTERPRETATION AND RECURSIVE BELIEF

A. Preprocessing and Features

The pipeline rejects nonfinite or shorter-than-eight-sample signals, removes the mean, and optionally applies a fourth-

TABLE III
ARGUS RESPONSE FEATURES

Group	Features
Amplitude	RMS, peak, crest factor
Temporal	zero-crossing rate, envelope-peak time, $1/e$ decay time
Spectral shape	centroid, bandwidth, 85% rolloff, dominant frequency, normalized entropy
Spectral allocation	low-, mid-, and high-band energy fractions
Quality	RMS-to-robust-noise SNR in dB

order zero-phase Butterworth bandpass from 250 Hz to $\min(7000, 0.96 f_s/2)$ Hz. Peak normalization is optional and disabled for feature extraction so amplitude remains informative. A Hann-windowed real FFT produces a unit-sum spectral energy distribution. Welch PSD, a short-time spectrogram, and a Hilbert envelope support inspection [20], [21].

Table III lists the 15 learned forward-response targets. The runtime inspector additionally returns a robust noise estimate. Low-, mid-, and high-band fractions span 250–1500, 1500–3500, and 3500–7000 Hz (clipped by Nyquist). Spectral entropy is normalized by the log of the number of FFT bins. Decay time is the first post-envelope-peak sample below $1/e$ of the peak. The noise scale is 1.4826 times the median absolute deviation (MAD) in the first 10% of the record.

B. Matched-Filter Spatial Likelihood

The inference engine subtracts the predicted healthy direct response, giving residual $v = y - \hat{y}_0$. It correlates v with the excitation using an FFT implementation, retains nonnegative lags, takes absolute magnitude, and applies a seven-sample maximum filter. Correlation is the classical matched-filter operation for a known waveform in noise [6].

For every grid-cell center j, the model predicts the bistatic delay

$$\hat{\tau}_j = t_0 + \frac{d(\boldsymbol{s}, \boldsymbol{z}_j) + d(\boldsymbol{z}_j, \boldsymbol{r})}{c}. \tag{22}$$

The correlation value at the nearest nonnegative lag is assigned to that cell, smoothed spatially by a Gaussian with $\sigma = 0.55$ cells, and robustly scaled between its 8th and 98th percentiles to $\tilde{C}_j \in [0, 1]$. The likelihood is

$$L(j) \propto \exp(T_{\text{eff}} \tilde{C}_j), \tag{23}$$

$$T_{\text{eff}} = 4.8 \,\text{clip}\left(\frac{\text{SNR} - 0.75}{1.8}, 0.30, 1\right), \tag{24}$$

where SNR is residual RMS divided by residual MAD scale. Scaling L to have mean one makes the Bayesian update numerically convenient without altering Eq. (10).

C. Estimation, Confidence, and Termination

ARGUS reports both the maximum-a-posteriori (MAP) grid center and posterior mean. The 2×2 covariance is the posterior-weighted covariance of cell coordinates and yields a confidence ellipse. Let m_{local} be posterior mass in a square neighborhood of radius $\max(1, \lfloor G/16 \rfloor)$ around the MAP cell. The displayed confidence is the bounded heuristic

$$Q = \text{clip}\{0.55 m_{\text{local}} + 0.45[1 - \bar{H}(P)], 0, 1\}. \tag{25}$$

The current stopping engine distinguishes budget exhaustion, OOD abstention, low modeled value, and a verified concentration stop. The latter requires the latest accepted verification to pass channel, OOD, diversity, and prior-location consistency checks, together with a configured concentration/area, entropy/value, or heuristic-risk criterion. Confidence or entropy alone cannot authorize a confident stop. A budget stop establishes no correctness. The historical baseline study used a ten-measurement budget; the default runtime budget is twelve.

VII. ADAPTIVE PHYSICAL EXPERIMENT PLANNER

A. Candidate Construction

The baseline generator mixes eight perimeter locations with up to eight probability-aware, spatially separated cell representatives as candidate sources. Receivers are the four perimeter corners plus the center. It cycles through bands 1.2–3.0, 2.2–4.4, and 3.4–6.2 kHz, with chirp, impulse, and chirp waveforms, amplitudes 0.42, 0.48, or 0.54, and 0.12 s duration. Near-coincident pairs are redirected and parameter duplicates removed, targeting the default pool of 48 candidates. The fast NEO path expands this baseline with finite waveform variants before applying constraints.

B. Counterfactual Hypothesis Disagreement

Let $\mathcal{K}$ contain up to $K = 24$ probability-aware spatial representatives in the baseline scorer, with represented posterior mass $M = \sum_{i \in \mathcal{K}} P(i)$ and normalized weights $w_i = P(i)/M$. The fast NEO feature path uses up to five representatives; regional arrival modes preserve all mass through regional assignments. For candidate e, the baseline forward signature is

$$\boldsymbol{g}_i(e) = \left[\tau_i, \log(A_i + 10^{-8}), \sin\varphi_i, \cos\varphi_i\right], \quad \varphi_i = 2\pi f_c \tau_i, \tag{26}$$

where $A_i = 0.65 e^{-\alpha d_i}/(1 + 5 d_i)$. Delay and log-gain dimensions are divided by 0.18 ms and 0.55, respectively, while sine/cosine encode circular phase without wrap discontinuities. With scaled signature $\tilde{\boldsymbol{g}}$, pairwise distinguishability and disagreement are

$$D_{ij}(e) = 1 - \exp\left[-\tfrac{1}{2}\|\tilde{\boldsymbol{g}}_i(e) - \tilde{\boldsymbol{g}}_j(e)\|_2^2\right], \tag{27}$$

$$D(e) = \sum_{i,j \in \mathcal{K}} w_i w_j D_{ij}(e). \tag{28}$$

The bounded information-gain proxy is

$$I(e) = H(P) M D(e), \tag{29}$$

in bits. It grows with current uncertainty, posterior mass represented by the leading hypotheses, and predicted separability. It is not the expected value of Eq. (11) under a calibrated observation distribution.

Algorithm 1: ARGUS closed-loop localization

Input: panel model, stopping thresholds, budget B.
Set $P_0(j) \leftarrow 1/G^2$ and $\mathcal{D} \leftarrow \emptyset$.
for $t = 0, \dots, B-1$ **do**
  Form sources from perimeter and leading cells.
  Generate and deduplicate candidate experiments.
  Retain top-K posterior hypotheses.
  for each candidate e **do**
    Predict $\{\boldsymbol{g}_i(e)\}_{i=1}^{K}$.
    Compute $I(e), D(e), U(e), C(e), R(e)$.
    $S(e) \leftarrow I + .35D + .22U - .16C - .22R$.
  end for
  Set $e_{t+1} \leftarrow \arg\max_e S(e)$ and execute it.
  Acquire y_{t+1} and construct L_{t+1}.
  Gate quality/channel/OOD; fuse $P_t \odot L_{t+1}^{\beta_t}$.
  Persist signal, features, scores, and posterior.
  Evaluate verified, OOD, value, and budget stopping.
end for
Return: MAP, mean, covariance, confidence, history.

Candidate coverage is

$$U(e) = \sum_{i \in \mathcal{K}} w_i \exp[-2.2 \min(d_{si}, d_{ir})]. \tag{30}$$

The implemented physical cost is

$$C(e) = 0.52a^2 \frac{d}{0.12} + 0.48 \frac{\|\boldsymbol{s} - \boldsymbol{s}_t\|_2}{\sqrt{2}}, \tag{31}$$

with zero motion term for the first experiment. Redundancy is the closest match to any previous experiment:

$$R(e) = \max_{h<t} \exp[-6\|\boldsymbol{s} - \boldsymbol{s}_h\|_2] \exp[-2|f_c - f_{c,h}|/3000]. \tag{32}$$

The final score is

$$S(e) = I(e) + 0.35D(e) + 0.22U(e) - 0.16C(e) - 0.22R(e). \tag{33}$$

The highest-scoring candidate is selected. The interface exposes all terms for the top five candidates, preventing the planner from becoming an opaque recommendation engine.

C. Programmatic Explanation

The selected recommendation receives a deterministic rationale. High disagreement produces a competing-regions explanation; broad entropy produces a coverage explanation; otherwise the system describes local ambiguity resolution. If repetition is below 0.25 after prior measurements, the explanation notes that the geometry and frequency avoid recent experiments. No language model is required, and every statement traces to planner state.

VIII. Trust-Gated Counterfactual Dual Control

A. Block-Structured Spatial and Metrology State

The original spatial grid is retained for location inference, while NEO stores a separate metrology block. The executed state can be written as

$$B_t = \{P_t(x, y), \hat{\boldsymbol{\eta}}_t, C_t, \rho_t, o_t\}, \tag{34}$$

where $\boldsymbol{\eta}$ contains wave velocity, attenuation, timing offset, noise scale, gain, source/receiver coupling, and source/receiver pose error. C_t retains velocity–timing covariance; other metrology quantities remain approximate marginal moments. This is not a stored hierarchical joint structural–metrology posterior. A seeded correlated particle-sampling helper exists and is tested, but the live controller does not call it. Size, severity, and type remain fixed priors.

Accepted diagnostic evidence updates location and may update a passive noise estimate. Reference calibration updates only supported metrology parameters; it never supplies structural evidence. Estimated velocity, timing, and noise means are synchronized into inference without changing hidden acquisition physics. Signal quality uses amplitude, clipping, finite-sample validity, repeat consistency, and placement proxies to produce $w_t \in [0, 1]$. The recursive structural update is

$$P_{t+1}(j) \propto P_t(j) \max\{L(y_{t+1} \mid j, e_{t+1}, \hat{\boldsymbol{\eta}}_t), \varepsilon\}^{\beta_t}, \tag{35}$$

where accepted quality has base weight $\max(w_t, w_{\min})$. A degraded channel and an OOD caution each multiply that weight by 0.35; an unreliable channel, rejected waveform, OOD abstention, or reference-only action sets the structural exponent to zero. There is no additional executed ρ_t/T_t multiplier. Model trust affects diagnostics, planning, and reported confidence. Rejected or unsupported observations cannot sharpen belief.

B. Dual Control: Diagnose or Calibrate

An action may interrogate the object or reduce uncertainty in the measuring process itself. Let U_s denote structural uncertainty, U_m combined metrology uncertainty, ρ model trust, and o the OOD score. NEO compares

$$V_{\text{cal}} = [U_m(0.70 + 0.30(1 - \rho)) + 0.35o]\, d_n, \tag{36}$$

$$V_{\text{diag}} = \hat{I}(e) U_s \rho, \tag{37}$$

with diminishing-return factor $d_n = (1 + 0.65 n_{\text{cal}})^{-1}$. Calibration is selected when $V_{\text{cal}} - V_{\text{diag}} > 0.05$, or when an explicit threshold/OOD rule requires it. The dominant nuisance term maps to an executable direct-path, coupling-repeat, phone-pose, microphone-level, or frequency-sweep action. Otherwise NEO selects diagnostic, exploration, or pre-stop verification. This is a transparent heuristic dual controller whose components and switching margin are returned by the API.

C. Counterfactual Observation Distributions and Bayes Risk

For the leading hypotheses h_i, the fast controller supplies compact feature predictions and assumed variances. Standardized mean separation and predictive variance yield average and worst-pair proxies. The legacy risk index is explicitly parameterized:

$$R_t = 12 p_{\text{miss}} + 3 p_{\text{false alarm}}, \quad p_{\text{miss}} = \text{clip}(1 - c_t + 0.45 o_t), \tag{38}$$

The symbols p_{miss} and $p_{\text{false alarm}}$ are heuristic indices, not fitted event probabilities. Predicted reduction is a bounded

function of information, separation, and trust. The default multiobjective utility is

$$\begin{aligned} J(e) = I(e) &+ 0.35D(e) + 0.55\Delta R(e) \\ &+ 0.65V_{\mathrm{cal}}(e) + 0.30\rho + \beta U(e) \\ &- 0.16C(e) - 0.22R(e) - 0.08T(e) - 0.08E(e), \end{aligned} \tag{39}$$

where β depends on exploration state. Alternative objectives expose information gain, Bayes risk, worst-case ambiguity, and information per physical cost. False-negative, false-positive, escalation, abstention, movement, time, and excitation losses are configuration values; they are research utilities, not certified maintenance economics.

D. Formal TG-CDI Policy

The compact implementation above is summarized by the following policy abstraction. Let $\mathcal{H}_t = \{h_1, \ldots, h_K\}$ contain spatially separated posterior representatives. At either live model level $\ell \in \{0, 1\}$, the fast controller uses a compact Gaussian feature approximation

$$q_{i,e}^{(\ell)}(y) \approx \mathcal{N}\left(\mu_{i,e}^{(\ell)}, \Sigma_{i,e}^{(\ell)}\right), \tag{40}$$

whose moments depend on geometry, frequency, and selected metrology moments. Amplitude, duration, and code effects enter separate waveform/cost heuristics rather than a validated full-waveform predictive distribution. Separation combines a bounded symmetric-KL transform, Bhattacharyya separation, and predictive-variance proxies. The legacy-named "Jensen–Shannon" score is $1-\exp(-\mathrm{SKL}/4)$, not actual JS divergence. The hard-rival term

$$S_{\min}(e) = \min_{i \neq j} S\left(q_{i,e}^{(\ell)}, q_{j,e}^{(\ell)}\right) \tag{41}$$

penalizes a candidate whose weakest represented pair has low predicted separation; it does not guarantee identification of all true rival locations.

A more general prospective policy could compare diagnostic expected loss reduction G_z with metrology information gain G_η using

$$\begin{aligned} U_D(e) &= \rho_t[w_z G_z + w_a S_{\mathrm{avg}} + w_m S_{\min}] - C(e), \\ U_C(e) &= w_\eta G_\eta + w_\rho \mathbb{E}[\Delta\rho_t \mid e] - C(e), \end{aligned} \tag{42}$$

These prospective utilities are not the executed fast-controller calculation. The fast controller first chooses an action class by Eq. (37), then ranks feasible candidates using Eq. (39). An unrestricted common-value competition across all action classes would instead be

$$(a_t, e_t)^* = \arg \max_{(a,e) \in \mathcal{A}_{\mathrm{safe}}} \{U_D, U_C, U_V, U_E, U_{\mathrm{abstain}}\}. \tag{43}$$

This unrestricted optimization is not implemented. The optional physical-loss controller compares calibration and diagnosis within its finite beam and observation horizon; a diagnostic can be labelled verification when configured concentration conditions hold. Unsupported planning enters abstention. Actual structural fusion uses

$$P_{t+1}(j) \propto P_t(j) \max\{L_t(j), \varepsilon\}^{\beta_t}, \tag{44}$$

Algorithm 2: TG-CDI interrogation cycle

Maintain location belief, metrology block, and trust.
Extract separated live rival modes $\mathcal{H}_t$.
Generate finite feasible geometry/waveform candidates.
Predict $q_{i,e}^{(\ell)}$ for every rival and action.
Score average and hardest-rival separation and risk.
Score calibration, verification, exploration, and cost.
Execute the highest-supported safe action or abstain.
Gate waveform/channel/OOD before evidence fusion.
Temper the belief update; update calibration/trust state.
Bind raw-data hash, model, rationale, and transition.
Stop, verify, escalate, or repeat under decision loss.

Fig. 2. Implemented TG-CDI research sequence at a high level. The fast and optional physical-loss controllers use different declared approximations.

with the explicit quality/channel/OOD gate described above, rather than a joint-posterior or direct trust-exponent update. Calibration-only actions set the structural exponent to zero while updating supported metrology state. An accepted packet is therefore not automatically structural evidence.

E. Joint Geometry–Waveform and Horizon Search

Source position, receiver position, band, amplitude, duration, and waveform are selected from finite candidates under hard limits. The library supports impulse, sine, chirp, tone burst, Ricker, multisine, phase-coded, complementary-coded, and spectrally notched excitation. Generated waveforms are bounded after modulation/coding. Frequency, amplitude, duration, spacing, no-go, and availability violations receive machine-readable reasons. Only cached analytical level 0 and physics-signature level 1 are live; neither is a validated FE or learned-surrogate adapter. The fast beam reranker scores geometric routes without observation or belief rollouts, with a configured nominal horizon of up to three. Actual observation branching belongs to the optional physical-loss controller and has horizon at most two. Neither search is globally optimal.

F. Discrepancy, OOD, and Decision Gating

An online ridge discrepancy model learns residual feature correction between predicted and observed responses and converts residual error to model trust $\rho \in [0, 1]$. Robust residual distance, an ensemble disagreement score, and a conformal tail score are combined for OOD monitoring; the conformal component remains withheld until ten reference scores exist. Reported decision confidence is capped by OOD evidence. The stopping controller considers confidence, normalized entropy, credible-region area, Bayes risk, expected value of another measurement, budget, and model trust. It may continue, verify, escalate to a reference method, or abstain. These mechanisms reduce unsupported certainty but do not guarantee statistical calibration under physical domain shift.

G. Evidence, Replay, and Distributed Operation

Every state-changing event can be recorded as canonical JSON with timestamp, prior hash, and SHA-256 entry hash. Exported ZIP bundles contain a manifest, session state, ledger, responses, planner explanations, configuration, versions, and

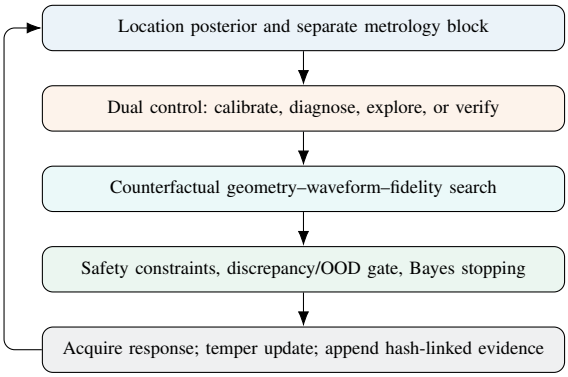


Fig. 3. ARGUS NEO closes inference, instrument calibration, experiment design, safety, and provenance in one recursive loop.

checksums; import verifies integrity before accepting the bundle. Replay readers support NPZ, CSV, and WAV response banks, while hidden truth remains sealed until the policy has stopped. SQLite-backed research jobs provide progress, cancellation, and interrupted-job recovery for counterfactual-bank generation, calibration, benchmarks, ablations, and surrogate studies. A phone progressive web application supplies camera overlay, inertial-quality metadata, and microphone acquisition; a reconnecting laptop edge client can register, heartbeat, poll, and return microphone measurements over a private LAN. These paths remain operator-supervised research tooling.

H. ARGUS-X Runtime Assurance and Failure Disposition

The extended interface retains normalized heuristic screening weights under legacy healthy/no-detectable-damage, known-damage-candidate, and unknown/unsupported labels. Let $p_d = \sigma(\ell_d)$ summarize residual-SNR evidence that passed structural gates; unsupported weight uses $\max(o, 1-\rho)$ before renormalization. These values are not fitted class probabilities, defect-existence inference, or POD estimates. Spatial modes are competing single-scatterer explanations. The legacy defect-count display is an ambiguity proxy, not inference of zero, one, or several actual defects.

Each sensor channel maintains a beta-form quality-weighted health score

$$\hat{r}_j = \frac{\alpha_j}{\alpha_j + \beta_j}, \tag{45}$$

where accepted observations add soft quality evidence and rejections add failure evidence. This score is not a calibrated sensor-failure probability. Channel health is evaluated before structural or calibration updates; an accepted waveform from a still-unreliable channel cannot update either state. Degraded channels temper evidence and request redundancy; unreliable channels force abstention. Environment metadata provides drift warnings. Duplicate identifiers/payloads, unsupported units/bandwidth, invalid samples/windows, clipping, silence, and infeasible actions are rejected. Timestamp skew is a warning rather than independently measured synchronization evidence.

A persisted emergency latch blocks acquisition until a human supplies an acknowledgment and reason; both transitions are audited. ARGUS may continue, reacquire/repair, mark a candidate region for reference inspection, or require a human. It cannot release a structure as healthy: a low residual-screening index instead requests independent reference verification.

The earlier 1,331-item ARGUS-X register recorded 277 narrow controls as tested, 588 partially mitigated, 244 requiring physical validation, 45 requiring literature/legal review, and 177 as bounded. These historical dispositions are not 1,331 solved claims or current-suite verification counts; the dated release report records the October checks.

IX. Learned Forward Surrogate and Data

A. Domain-Randomized Synthetic Set

The optional learned component approximates expected response features, not the final posterior. Each row contains 24 inputs describing defect state, panel/material nuisance variables, and experiment parameters, and 15 outputs from Table III. Three thousand examples were generated with locations in $[0.08, 0.92]^2$, radii 0.04–0.14, severity 0.35–0.96, all four defect types, panel sizes 0.45–0.85 by 0.30–0.60 m, velocity 145–245 m/s, attenuation 1.0–2.4 m^{-1}, resonance 2.5–4.2 kHz, damping 70–135 s^{-1}, noise 0.003–0.014, and randomized probes, bands, amplitudes, and waveforms.

The PyTorch [22] network is

$$24 \to 96 \to 96 \to 15, \tag{46}$$

with SiLU activations and LayerNorm after the first hidden layer. Inputs and targets are standardized on training statistics. The historical training used Smooth L1 loss, AdamW at learning rate 1.8×10^{-3} and weight decay 10^{-4}, batch size 128, a seeded 70/15/15 split, and early stopping. Not all randomized damping, noise, and system-delay quantities appear in the 24 inputs, so the predictor must average over unobserved variability. The surrogate remains outside the live planner.

B. Measured LMSD Adapter

The downloader queries the official Zenodo record, downloads declared NPZ files, and verifies each published MD5 checksum. The adapter converts each complex 0–1600 Hz FRF to a 0.12 s impulse response, normalizes known node coordinates, encodes the added mass as a dense inclusion, and extracts the same feature contract. Six scenarios times seven sources times seven receivers yield 294 channels. To prevent leakage, splitting is by entire damage scenario: 196 training, 49 validation, and 49 test channels. These channels are correlated views of six physical damage configurations, not 294 independent defects.

C. Measured Acoustic and Paired Localization Bridges

A second checksum-verifying adapter processes the TU Darmstadt GFRP acoustic dataset [23]: 2,568 unaugmented microphone taps from eight 360 by 280 mm plates, including

2,100 intact and 468 defect-labelled point measurements. Signals are rationally resampled from 51.2 to 16 kHz and converted by the production feature contract. Splitting holds out complete physical plates. Separate healthy/damage robust CORAL maps align synthetic and measured first/second feature moments, while a robust empirical reference uses only gain-insensitive features. For external physical acquisitions, only the upper 10% measured-reference tail contributes to OOD caution or abstention; simulation sessions are unaffected.

The Bologna acoustic-emission record [24] adds 1,000 ray-traced Tx–Rx paths with fourth-order boundary reflections and 27 measured channels from nine known impact positions on a $1\times1\times0.003$ m aluminium plate. Its 2 MHz, 250 kHz domain is not mixed with the low-frequency GFRP waveforms. It instead supports a separate simulation-transfer timing benchmark and a measured few-shot localization calibration. Repository files are accepted only after publisher-declared size and MD5 verification, and manifests preserve record, DOI, license, and selected-file provenance.

X. Physical and Software Realization

A. Acquisition Abstraction

All devices implement connect, disconnect, acquire, and status operations. `SimulationDevice` executes Eq. (15); `MicrophoneDevice` records through the local audio stack; `UploadedSignalDevice` accepts WAV files; and `SerialProbeDevice` performs a line-framed transaction with an ESP32. Missing audio or serial hardware is reported without disabling simulation.

Uploads are limited to 10 MB, checked for allowed media types, decoded in memory, and explicitly downmixed to mono. Finite recordings must cover the declared experiment window and support the excitation band at their original sampling rate before polyphase resampling. Longer windows are cropped with recorded provenance; shorter recordings are rejected without padding missing evidence. Physical-unit inputs require declared conversion to the normalized inference contract. The upload API does not accept arbitrary filesystem paths. Simulation truth is omitted from ordinary payloads until reveal.

B. Calibration and Coordinates

The simulation calibration routine generates three healthy-reference acquisitions and records supported timing/velocity, noise, gain, and baseline-RMS diagnostics; it does not estimate resonance. Physical calibration requires acquired reference evidence. The session controller may request direct-path, coupling-repeat, phone-pose, microphone-level, or frequency-sweep actions. Each update requires its measured diagnostics; absolute pose requires an external pose reference, and waveform quality does not identify separate transmitter/receiver coupling. Supported metrology changes are persisted. Normalized coordinates provide a common interface, while predictions and reported millimetres use actual panel dimensions through Eq. (6).

A camera-guidance mode asks the operator to click the four panel corners in top-left, top-right, bottom-right, bottom-left order. Eight homography parameters are solved from the four point correspondences [25]. The browser overlays source, receiver, propagation path, posterior field, suspected region, and a 90% covariance ellipse. Visual-position error and inertial deviation are sent as quality metadata and reduce evidence weight; this remains approximate operator guidance, not certified optical metrology.

C. ESP32 Probe

The current classic-ESP32 reference uses DAC GPIO25, conditioned ADC GPIO34, GPIO26 as an excitation timing marker, and serial at at least 115200 baud for supported acquisition. Its version-2 experiment-bound protocol retains concurrent drive/capture, pretrigger samples, actual call timestamps, DAC codes and complete-frame status, as specified in the later acquisition section. Legacy excite-then-record firmware is refused by arrival inference. The DAC must feed an appropriate AC-coupled amplifier rather than power the exciter directly. The receiver needs input protection, bias, buffering and antialias conditioning. Electrical timing and transfer remain unmeasured.

D. Persistence and Reproducibility

Additive SQLite migrations preserve legacy sessions while storing separate inference blocks, probes, events, jobs, model records, and hash-linked ledger entries, alongside configuration, calibration, compressed responses, features, scores, and posterior history. NumPy/SciPy provide vectorized numerics and signal processing [26], [27]. Fixed seeds cover defect generation, observation noise, splits, training, and paired policy comparison. Full-precision structural state retains small supported hypotheses; rejected observations do not become accepted repeat references after hydration. Source fingerprints and saved outputs permit checks against the actual changed source, rather than only a repository commit identifier.

XI. Evaluation Protocol

The evaluation addresses five questions:

- **RQ1:** Does active planning concentrate spatial belief faster than random or uniform probing?
- **RQ2:** Does lower uncertainty translate into lower localization error under the same budget?
- **RQ3:** Can a small learned surrogate reproduce domain-randomized signal features?
- **RQ4:** Do those learned features transfer across held-out physical damage scenarios?
- **RQ5:** What experiment would be required to establish the proposed TG-CDI technical effect?

For RQ1–RQ2, 30 medium-preset defects were generated with seeds 100–129. Each of three policies operated on the identical hidden defect for a maximum of ten measurements. *Random* sampled continuous source and receiver positions and a frequency band. *Uniform grid* followed a fixed geometry sequence. *ARGUS* used Eq. (33). Policies were compared using

mean and median localization error, normalized final entropy, entropy reduction, entropy area under the measurement curve (AUC), physical cost, experiment count, and success within 10, 15, 20, and 30 mm. Paired bootstrap 95% intervals were computed over the shared cases. No assertion required error to improve monotonically.

For RQ3, the 3000-example synthetic model used seed 23 and independent random splitting. For RQ4, the LMSD model used the same architecture and seed but held out complete scenarios. Smooth L1 values across these two data regimes are not directly comparable because their target statistics and split difficulty differ.

RQ5 is a prospective, preregistered test rather than a result in this paper. On a complete physical response bank, compare TG-CDI against raster coverage, a uniform sequence, random selection, fixed-network reconstruction, entropy-only active selection, calibration-disabled TG-CDI, trust-disabled TG-CDI, and worst-rival-disabled TG-CDI. Freeze losses and stopping thresholds before revealing blind truth. Primary endpoints should be measurements and physical cost to reach a fixed decision-risk threshold; secondary endpoints should include localization error, probability of detection with confidence bounds, false-stop rate, abstention rate, calibration coverage, remount repeatability, temperature-shift robustness, and wall-clock time. The differentiating hypothesis survives only if the full policy improves the prespecified primary endpoint without an unacceptable false-negative or unsupported-stop penalty.

XII. Results

A. Planner Benchmark

Table IV reports the saved benchmark, not hand-entered target values. ARGUS achieved the lowest mean error, final entropy, entropy AUC, and mean experiment count. At 15 mm it localized 73.3% of cases, versus 56.7% for random and 60.0% for the grid. At 20 mm all methods reached 93.3%; therefore the strongest separation occurs at the tighter 15 mm criterion.

Relative to random probing, ARGUS lowered final entropy by 15.9%, lowered mean error by 7.8%, and used 2.9% less modeled cost. Relative to the grid, it lowered final entropy by 32.5% and mean error by 7.0%, but used 4.3% more modeled cost. The result illustrates Eq. (1): information efficiency and physical cost are related but distinct objectives.

Figure 4 shows mean trajectories. ARGUS does not dominate localization error at every intermediate step; its mean error is especially high after the second measurement and briefly rises again after the sixth. This is permissible in a noisy recursive estimator and reinforces why entropy and error must both be reported.

Paired results in Table V qualify the aggregate means. ARGUS reduced final entropy in 28/30 cases versus random and 30/30 versus grid, with both confidence intervals above zero. In contrast, both mean-error intervals include zero and strict error win rate is 30%. Thus RQ1 is supported in this simulator, while RQ2 remains suggestive rather than statistically established. Tied MAP cells and a few large improvements explain why mean error can improve despite a low strict-win fraction.

B. Historical NEO Engineering Diagnostics

The following saved diagnostics precede the 20 September audit corrections. They are retained for provenance, not current-version validation. The historical ECE value below compared a heuristic score with a different 20 mm event and is not a valid probability-calibration certificate. Current reliability evaluation compares posterior mass within 20 mm with the corresponding localization event. Table VI preserves earlier seeded engineering checks. They are separated from the 30-case baseline because the quick NEO matrix contains only two cases and a five-measurement budget. None of nine policies met the then-configured joint stopping criterion, all used five measurements, and therefore observed inspection compression was $1.0\times$. Full NEO mean error was 21.70 mm and success within the configured radius was 0.50; uniform, Bayes-risk-only, dual-control-only, and one-step horizon variants were 15.95 mm and 1.00 in these two cases. This historical pipeline smoke test supplies no basis for a superiority claim.

The historical mismatch demonstration altered effective velocity by 2% and attenuation by 4% under the same hidden truth, seed, and budget. NEO used eight diagnostic, three calibration, and six verification actions, ended at 14.65 mm error, model trust 0.517, and a caution state. The deliberately naive controller used 15 diagnostic and no calibration actions, ended at 157.54 mm with confidence 0.735, and triggered the false-confidence diagnostic. This one paired case illustrates the intended mechanism in that earlier implementation; it is neither an estimated effect size nor evidence of physical generalization. In four calibration smoke cases, nominal 50–95% region coverages were all 1.0. The accompanying 0.715 proxy-event mismatch score is deprecated, not a valid ECE estimate. Tiny-sample apparent coverage must not be advertised as calibration.

C. Historical September Synthetic Audit Evaluation

The September source version compared the optional physical-loss controller with the legacy proxy using six separate seeds (53000–53005), six measurements per run, twelve shared diagnostic candidate opportunities per step, and two acquisition contexts. All 24 runs completed without an execution failure; source fingerprints did not change during that run. The mismatched context used effective velocity multiplied by 1.02, attenuation by 1.04, and noise by 2. Truth was used for scoring, not action selection. Table VII retains both outcomes without discarding unfavourable cases. Subsequent October evidence and uncertainty changes prevent these values from serving as current-source validation.

The mean error advantage of the risk controller was 31.32 mm with a paired bootstrap 95% interval of $[-2.71, 93.38]$ mm in nominal conditions, and -53.56 mm with interval $[-174.99, 17.82]$ mm under mismatch. Both intervals cross zero. One legacy outlier strongly affects the nominal mean;

TABLE IV
PAIRED SIMULATION BENCHMARK: 30 MEDIUM-PRESET DEFECTS, TEN-EXPERIMENT BUDGET

Policy	Mean error (mm)	Median error (mm)	Mean exp.	Final $\bar{H}$	Entropy red.	Entropy AUC	Cost	≤ 15 mm (%)	≤ 20 mm (%)
Random	13.38	14.44	10.00	0.498	0.502	0.748	2.665	56.7	93.3
Uniform grid	13.26	13.84	10.00	0.621	0.379	0.818	**2.481**	60.0	93.3
ARGUS	**12.33**	**13.28**	**9.83**	**0.419**	**0.581**	**0.740**	2.588	**73.3**	93.3

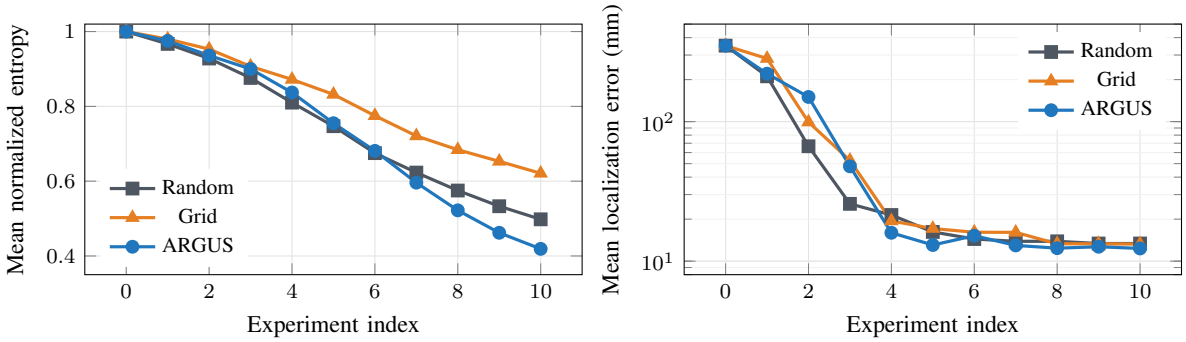


Fig. 4. Mean uncertainty (left) and localization-error (right, logarithmic axis) trajectories over the same 30 hidden defects. Lower is better.

TABLE V
PAIRED ARGUS ADVANTAGES (95% BOOTSTRAP CI)

Comparison	Mean error advantage (mm)	Mean entropy advantage	Win rate (error / entropy)
vs. random	1.048 $[-0.667, 2.812]$	0.079 $[0.063, 0.095]$	0.30 / 0.93
vs. grid	0.928 $[-1.021, 2.757]$	0.202 $[0.180, 0.224]$	0.30 / 1.00

TABLE VI
OBSERVED ARGUS NEO DIAGNOSTIC RUNS

Run	Cases	Key error	Key uncertainty	Interpretation
Nine-policy quick matrix	2	NEO 21.70 mm	$\bar{H} = 0.909$	No crite-rion/compression
Quick calibration	4	Old score 0.715	Coverage 1.0	Event mismatch; deprecated
Mismatch, seed 17	1 paired	14.65 vs. 157.54 mm	NEO caution/OOD	Mechanism demo only
Active surrogate	180	MAE 94.10→89.01	5.41% decrease	Synthetic oracle only

success within 20 mm was 5/6 for both controllers nominally and 0/6 for both under mismatch. Computation averaged about 21 seconds per six-measurement case for the risk controller versus 0.35–0.39 seconds for legacy on this machine. The risk controller remains disabled by default. This small study establishes neither superiority, calibration benefit, measurement compression nor acceptable real-world accuracy. The full artifact is `research_results/audit_risk_validation.json`.

D. Historical Forward-Surrogate Results

The following table records the earlier 3,000-sample checkpoint. A separate retained 15,000-sample checkpoint records validation loss 0.18470, test loss 0.19815, standardized MAE 0.45690, and mean feature $R^2 = 0.47567$. An additional 50,000-sample candidate and paired feature comparison are reported separately in `docs/SURROGATE_TRAINING_2026_09_20.md`; it does not replace a live-controller

TABLE VII
HISTORICAL SEPTEMBER CHECK: SIX PAIRED SEEDS PER CONTEXT

Context	Legacy mean error	Risk mean error
Nominal	43.90 mm	12.58 mm
Mismatch + noise	150.22 mm	203.78 mm

model. These response-feature results do not establish defect-localization accuracy or a validated runtime surrogate. The synthetic surrogate stopped at epoch 27. Table VIII shows that amplitude, decay, and SNR-related features are learnable under domain randomization, while dominant frequency and envelope timing are poorly captured. Mean feature $R^2 = 0.409$ is useful as an engineering diagnostic but not sufficient justification to replace the physics signature in safety-relevant planning.

The measured LMSD scenario-held-out result is much weaker: mean $R^2 = 0.001$ and standardized MAE 0.449. High-band energy alone achieved $R^2 = 0.875$; many other features were negative out of sample. The adapter therefore demonstrates an honest data path and exposes domain shift rather than validating physical generalization. In particular, test Smooth L1 0.190 should not be interpreted as better than the synthetic 0.225 because the standardized targets and held-out units differ.

E. Measured-Domain Bridge Results

Table IX reports two scoped real-data checks. The GFRP detector uses four-fold GroupKFold with complete plates held out; repeated taps, sides, runs, and points from a held-out plate cannot enter training. Balanced accuracy was 0.775, ROC–AUC 0.911, average precision 0.699, F1 0.634, and Brier score

TABLE VIII
LEARNED FORWARD-RESPONSE EVALUATION

Data	Samples	Test Smooth L1	Std. MAE	Mean R^2	Split
Synthetic	3000	0.2245	0.5014	0.4092	seeded 70/15/15
LMSD measured	294	0.1902	0.4485	0.0010	held-out scenario

Synthetic best R^2: SNR 0.910, RMS 0.861, peak 0.825, crest 0.758, decay 0.622. Synthetic weak examples: dominant frequency -0.019, envelope-peak time 0.016.

TABLE IX
SCOPED MEASURED-DOMAIN BRIDGE RESULTS

Record/task	Split	Primary result	Scope limit
GFRP acoustic detection	held-out plate	Bal. acc. 0.775; AUC 0.911	pointwise taps
Al impact, simulation transfer	held-out position	Mean error 0.175 m	one plate
Al impact, measured few-shot	held-out position	Mean error 0.141 m	eight calibration points

0.097. These are pointwise screening results, not arbitrary multistatic localization.

For the aluminium impact record, direct simulation transfer produced 0.175 m mean localization error. A three-neighbour measured localizer trained on eight positions and tested on the ninth reduced the leave-one-position-out mean to 0.141 m, or 19.6%. An affine timing correction increased the mean to 0.197 m and was therefore rejected by a negative-transfer guard. The improvement demonstrates the value of a small measured calibration set, but one plate, nine interior positions, and transient impacts cannot validate persistent hidden-defect localization.

F. Software Verification

The dated `docs/RELEASE_READINESS_REPORT.md` records final executed test counts, commands, frontend checks, and any outstanding failures. Historical audit checks include rank-one calibration information, different-distance identifiability, preservation of unobserved coupling uncertainty, OOD-before-fusion, stale-verification rejection, anchored detection of rehashed history, and stereo/unsigned PCM normalization. HTTP/API checks do not substitute for visual browser inspection or physical-device validation. A nonfatal test-client deprecation warning is not an execution failure.

G. October Security and Scientific Reproducibility Hardening

The 7 October revision adds regression cases for accepted waveforms from unreliable channels, calibration without measured support, rejected repeat references, finite mono recordings, declared acquisition windows, original recording Nyquist support, and normalization of declared physical units before inference. Server-owned acquisition provenance selects physical reference checks; a client sensor name cannot disable them. Stable probability updates reject invalid support, retain small supported hypotheses across persistence, and keep unlearned attribute priors out of location-uncertainty scoring. Amplitude/duration changes or reciprocal source–receiver order alone do not count as diverse verification.

Network regression cases exercise loopback-only defaults, explicit authenticated LAN mode, host/origin checks, bounded requests/WebSocket frames, work/concurrency limits, and emergency-stop access. Artifact regressions exercise bounded archive/array inspection, escaping paths, compressed payload limits, metadata validation, planner/ledger consistency, truth-sealed replay, and source fingerprints that include dirty and untracked code. These controls reduce specified software failure paths; they are not a complete security proof, acquisition authentication, or physical-performance evidence.

TABLE X
OCTOBER RELEASE SYNTHETIC CHECK (EIGHT SEEDS PER CONDITION/POLICY)

Condition	Policy	Mean error (mm)	Success ≤ 20 mm
Nominal	Legacy proxy	16.04	0.750
Nominal	Nuisance-aware	14.30	0.750
Nominal	Risk calibration	13.35	0.750
Mismatch	Legacy proxy	189.95	0.000
Mismatch	Nuisance-aware	144.36	0.125
Mismatch	Risk calibration	143.71	0.125

H. 7 October Current-Source Engineering Validation

The fixed release protocol used seeds 64000–64007, six observations, twelve shared diagnostic opportunities, three policies, and nominal and mismatched/noisy conditions. All 48 runs completed; the pre/post source hashes agree and match the reviewed backend files. This is a small synthetic engineering check, not a powered physical trial. Table X retains every outcome. Nuisance-aware error advantages over legacy were 1.74 mm with paired bootstrap 95% interval $[-6.78, 9.50]$ mm nominally and 45.59 mm with interval $[-22.85, 107.10]$ mm under mismatch. Risk-controller advantages were 2.69 mm with interval $[-0.78, 7.59]$ mm and 46.24 mm with interval $[-10.48, 112.27]$ mm, respectively. Every interval crosses zero. All policies succeeded within 20 mm in 6/8 nominal cases. Under mismatch, legacy succeeded in 0/8 and both optional policies in 1/8. Mismatch failures remain substantial; the software guards do not establish physical robustness. Timing on the shared workstation is descriptive. The artifact is `research_results/release_validation_2026_10_07.json`; the fixed protocol is `docs/VALIDATION_PROTOCOL_2026_10_07.md`. No seed filtering, post-result tuning, physical relabeling or superiority claim is made.

XIII. TECHNICAL DIFFERENTIATION AND CANDIDATE PROTECTABLE NUCLEUS

The system-level research hypothesis is that a controller able to improve both object and instrument estimates may use a measurement budget more effectively. This effect remains unproven. Seven implemented aspects are relevant to testing that hypothesis:

1) **Live rival construction:** separated modes are extracted from the current posterior, so the next experiment targets the ambiguity that actually remains.
2) **Joint actionable uncertainty:** structural and metrology marginals determine the next action, rather than being passive confidence displays.

3) **Action arbitration:** the fast policy uses declared switching heuristics; the optional physical-loss policy compares reference calibration and diagnosis under one modeled localization loss.
4) **Finite candidate selection:** geometry/frequency predictions discriminate represented rivals, while waveform, amplitude, duration, and costs also influence heuristic ranking. The two live feature-model levels and the optional scalar-arrival planner have separate approximation boundaries.
5) **Trust-gated evidence:** quality, channel reliability, model discrepancy, and OOD evidence control the strength of the physical likelihood instead of merely changing a displayed confidence score.
6) **Constraint-aware execution:** infeasible frequency, amplitude, spacing, no-go, and unavailable-device actions are rejected and explained before execution.
7) **End-to-end auditability:** decomposed alternatives, deterministic explanations, raw-response checksums, posterior evolution, model identity, and decisions form a verifiable evidence chain.

The candidate subject for professional examination is an apparatus-limited acoustic sequence that maintains a location belief and correlated timing/velocity metrology block, compares direct-path calibration with diagnostic bistatic actions under a common modeled localization loss including a subsequent diagnostic observation, executes one feasible finite-bank action, gates measured evidence before structural fusion, and requires fresh consistent verification for a confident stop. It also records action, observation, rationale, posterior transition, and model identity. This supported sequence is narrower than a general joint posterior, hardest-rival optimal policy, arbitrary waveform design, or multi-fidelity physical solver. Neither the established mathematical components nor their combination are presumed patentable.

Table XI is a technical differentiation map, not a legal claim chart. It shows why the argument must rest on the ordered interaction among elements rather than ownership of established components.

This positioning does not establish novelty or non-obviousness. The reviewed references already cover important subsets, and additional patents may be closer. A practitioner could also argue that combining Bayesian design, calibration, robust updating, and audit logging is obvious. The counterargument must identify a concrete technical interaction and measured effect—for example, that hardest-rival planning plus calibration arbitration and trust-tempered evidence reaches a prespecified decision risk with fewer safe physical interrogations under sensor/model shift. Only a professional, jurisdiction-specific search, element-by-element chart, inventorship review, and prosecution strategy can determine patentability, eligible subject matter, and freedom to operate.

Commercially, TG-CDI is best positioned as adaptive inspection-orchestration middleware for low-channel, movable-probe, measurement-budget-limited, or access-constrained workflows. It could complement a robot, phased-array instrument, guided-wave tool, permanent SHM network, or acoustic-emission service rather than claim to replace their qualified acquisition hardware. The sellable proposition is a measurable reduction in inspection actions or decision cost at matched detection/false-alarm risk, plus calibrated abstention and replayable evidence. That proposition remains prospective until the blind physical study in Section XV is complete.

XIV. Limitations and Threats to Validity

A. Model and Inference Limits

The simulator is nondispersive and uses one effective velocity. A real composite plate supports multiple dispersive, anisotropic modes and boundary reflections. The location grid is conditional on one dominant scatterer; size, severity, and type remain unchanged priors, and healthy/existence/count reconstruction is absent. Baseline subtraction assumes sufficient nominal-twin calibration. Successive generalized likelihood multiplication approximates conditional independence; shared error can still over-concentrate the belief. The correlated particle helper is not a deployed hierarchical inference engine.

Confidence in Eq. (25) is a heuristic mixture, not a calibrated posterior coverage probability. The historical ECE compared that proxy with a different event and has been deprecated; a corrected event-matched reliability calculation still requires independent physical validation. The planner utilities and cost conversion were engineered rather than elicited from a certified maintenance process. The conformal tail is withheld before ten scores and is only meaningful if future calibration scores are exchangeable with deployment. The block-structured state, locally Gaussian propagation covariance, compact counterfactual features, online ridge correction, and bounded lookahead are deliberate approximations.

B. Evaluation Limits

Thirty baseline synthetic cases provide a useful paired study, not population-scale validation. The baseline mean-error intervals cross zero; no claim of superior physical localization is warranted. The expanded NEO matrix has only two cases and is exclusively a software/research smoke test. Baselines do not include every advanced tomography, Gaussian-process design, reinforcement-learning, or optimal-array method. Hyperparameters were not evaluated in a nested holdout, and most studies share simulator assumptions with inference.

The LMSD experiment has only six independent damage scenarios, several with multiple masses. Hammer/accelerometer FRFs at 0–1600 Hz differ from the prototype's 1.2–6.2 kHz chirp interrogation. Added masses are scattering surrogates, not cavities or delaminations. Scenario-held-out performance correctly reveals that correlated channel count cannot replace independent physical diversity.

C. Operational Limits

Microphone measurements are vulnerable to airborne sound and ambient noise. Probe remounting changes coupling. Camera homography assumes a planar stationary panel and accurate clicks; inertial/visual metadata are quality proxies, not

TABLE XI
CLAIM-ORIENTED TECHNICAL DIFFERENTIATION MAP (NOT A PATENTABILITY OPINION)

Element	Crowded category	TG-CDI-specific coupling proposed for examination	Intended technical effect / evidence still required
Separate belief blocks	Bayesian localization and nuisance estimation [5]	Location and metrology moments are actionable; timing/velocity covariance is retained	Avoid diagnosing an instrument error as damage; prove on calibration-shift specimens
Live rivals	Multi-path prediction and guided-wave localization [11]	Posterior modes generate candidate-specific response distributions; score both average and worst-pair separation	Direct the next shot at the unresolved ambiguity; prove fewer shots at fixed decision risk
Action arbitration	Bayesian placement, active calibration, digital twins [4], [7], [13]	Optional common-loss reference/diagnostic comparison; other action classes use declared heuristics and stop rules	Spend a scarce measurement on object or instrument as warranted; ablate each branch
Finite physical selection	Phased arrays, fixed networks, waveform imaging [10], [12]	Finite bistatic/waveform candidates under access/cost limits; live analytical feature levels only	Reduce placements for constrained inspection; prove against raster, fixed-grid, and entropy-only policies
Trust-tempered update	Robust inference and OOD detection	Quality, accumulated channel reliability, model discrepancy, and OOD determine a likelihood exponent, including zero	Prevent unsupported posterior sharpening; prove false-stop and coverage behavior under shift
Evidence binding	Industrial traceability and integrity logging	Alternative scores, executed physical settings, raw-response hash, model identity, and posterior transition share one causal record	Reproduce why a measurement was selected and how it changed belief; perform independent replay/tamper audit

traceable metrology. Clock-skew checks, heartbeats and reconnect logic do not provide hard real-time synchronization. The v2 ESP32 scheduling marker and ADC call timestamps require external electrical timing/transfer validation; analog conditioning and physical power limits are still apparatus requirements. Unaligned physical recordings carry zero inference weight. The prototype is not certified NDE equipment and must not be used as the sole basis for safety-critical maintenance decisions.

XV. PHYSICAL VALIDATION AND DATA-COLLECTION PLAN

A credible next study should use at least three nominally identical panels, eight repeatably mounted perimeter transducers, an instrumented exciter, five repeats per condition, and healthy measurements before and after each damage block. Reversible bonded masses, damping patches, or paired magnets are appropriate for commissioning. True cavities or delaminations require manufactured specimens—for example flat-bottom holes or embedded release films—and independent truth from ultrasound, radiography, or destructive sectioning where justified.

For the current acquisition contract, collect all 56 ordered non-self source–receiver pairs, three frequency bands, and five repeats. One physical state yields $56 \times 3 \times 5 = 840$ waveforms. Twenty known locations plus healthy references yield approximately 17,000 signals. Preserve raw waveforms and record specimen, material, thickness, defect geometry and truth method, source and receiver coordinates, excitation, repeat, temperature, humidity, coupling force, remount identifier, calibration, timestamp, and code version.

Split development data by complete defect location, stronger validation by complete specimen, and the final test by a blind panel whose truth remains hidden until predictions and stopping decisions are frozen. Offline planner evaluation additionally needs a complete counterfactual bank: every admissible experiment must be recorded for every hidden state so all policies query the same physical response table. Report error versus measurement budget, entropy calibration, success radii, abstention, remount robustness, and temperature shift.

XVI. REPRODUCIBILITY

The repository provides commands for the complete workflow:

```
python run_argus.py
python run_edge_node.py --help
python scripts/run_neo_demo.py
  model_mismatch
python scripts/neo_benchmark.py --cases 2
python scripts/neo_benchmark.py --ablation
python scripts/neo_calibration.py --cases 4
python scripts/generate_counterfactual_bank.py
  tiny
python scripts/active_learn_surrogate.py
  --cases 180
python scripts/demo_simulation.py
python scripts/evaluate_model.py
python scripts/generate_dataset.py
python scripts/train_model.py
python scripts/download_lmsd_dataset.py
python scripts/prepare_lmsd_dataset.py
pytest
cd frontend
npm test
npm run lint
npm run build
```

The externally downloaded data are intentionally excluded from version control; DOI, license, source filenames, and checksums are preserved. Trained metadata records seeds, split units, sample counts, feature names, per-feature MAE, and R^2. Benchmark JSON preserves per-case trajectories and summary values. Evidence exports independently preserve the effective configuration, versions, ledger chain, planner alternatives, and response checksums. Every result in the NEO diagnostic table is labeled simulated unless a measured source is explicitly identified.

XVII. RESEARCH EXTENSION: NUISANCE-MARGINAL ARRIVAL-TIME DESIGN

This extension, implemented after the earlier studies, replaces the diagnostic information proxy with a numerically integrated information calculation for an explicit, approximate

observation model. It is an opt-in research mode, not an efficacy claim. The established planner remains the default.

The 80-run study below belongs to the pre-audit source version. The September revision added velocity–timing covariance to the arrival variance; October changes additionally altered evidence gating, support metadata, and numerical contracts. The older results must not be attributed to either changed model. The separate physical-loss controller and its historical September 24-run evaluation are described in the main text. The October source requires its own completed, source-fingerprinted evaluation artifact.

A. Prior Art and Scope

Nuisance-aware Bayesian experimental design is established: Bartuska et al. derive small-noise approximations for nuisance marginalization [28]. Robust information design under prior ambiguity is also established [29]; ARGUS does not implement that criterion or inherit its guarantees. In the directly adjacent acoustic domain, Raorane et al. optimize sensor-cluster placement for acoustic-emission source localization in isotropic plates and include physical validation [30]. Consequently, acoustic localization combined with information-based sensing cannot be claimed as a new principle here. The narrower contribution is an inspectable implementation of live posterior-region compression, nuisance-marginal scalar design, numerical diagnostics and paired evaluation within the sequential bistatic-interrogation workflow. Its legal novelty and commercial superiority have not been established.

B. Mass-Preserving Spatial Compression

Let p_i denote posterior mass at cell position $\boldsymbol{x}_i$, expressed in physical metres. Up to five spatial representatives define nearest-centre regions R_k using physical Euclidean distance. Unlike renormalizing only the representative cells, the regional weights preserve the entire posterior:

$$w_k = \sum_{i \in R_k} p_i, \qquad \sum_k w_k = 1, \qquad q_{i|k} = p_i / w_k. \tag{47}$$

For source $\boldsymbol{s}$ and receiver $\boldsymbol{r}$, define $L_i = \|\boldsymbol{x}_i - \boldsymbol{s}\| + \|\boldsymbol{x}_i - \boldsymbol{r}\|$. The region-conditioned expectation and variance of this bistatic path are computed using $q_{i|k}$. With nuisance posterior means b for timing offset and v for velocity, the approximate regional arrival distribution is

$$Y \mid K = k, e \sim \mathcal{N}(\mu_k, \sigma_k^2), \tag{48}$$

$$\mu_k = b + \mathbb{E}_k[L]/v, \tag{49}$$

$$\begin{aligned} \sigma_k^2 = {} & \mathrm{Var}_k[L]/v^2 + \sigma_{\mathrm{meas}}^2 + \sigma_b^2 \\ & + \mathbb{E}_k[L^2]\sigma_v^2/v^4 + \sigma_{\mathrm{pose},k}^2 \\ & - 2\mathbb{E}_k[L]\,\mathrm{Cov}(v,b)/v^2. \end{aligned} \tag{50}$$

The pose term propagates independent source/receiver placement second moments through first-order path gradients, scaled by the actual panel dimensions. Coincident probe/cell geometry uses a conservative finite sensitivity rather than an undefined derivative. A scalar pose bias without a known direction is included through its squared magnitude. These are assumptions, not measured pose-error distributions.

The measurement floor is $\sigma_{\mathrm{meas}}^2 = 1.5\times10^{-8} + (3/f_s)^2/3$ in seconds squared, combining the existing model floor with an approximate matched-filter sampling neighbourhood. It is not a learned timing calibration or a waveform-dependent precision bound. Moment matching and first-order nuisance propagation make this a surrogate for the real observation likelihood.

C. Information Calculation and Numerical Audit

For Gaussian density f_k, the design utility is

$$I(K; Y \mid e) = \sum_k w_k \int f_k(y) \log_2 \frac{f_k(y)}{\sum_j w_j f_j(y)}\, dy. \tag{51}$$

It targets the regional latent variable K, not exact cell location or the full waveform. Deterministic Gauss–Hermite quadrature uses 24 and 48 nodes, increasing to 96 when their difference exceeds 0.002 bits. Log-sum-exp evaluation avoids direct mixture underflow. The implementation reports raw information, the clamp to $[0, H(K)]$, node count and last-order difference. That difference is a numerical diagnostic, not a certified error bound. Independent adaptive integration tests check a nontrivial mixture; other tests check identical-response and separated-response limits, unit invariance, physical-coordinate partitioning and non-mutating planning.

For compatibility with existing multiobjective weights, the legacy-named API score becomes $I(K; Y \mid e)T/\max\{\log_2 K, 1\}$, where T is model trust. Unscaled information is exposed separately in bits. The remaining cost, waveform, repetition, constraint and baseline-guard terms are retained; this is not a pure mutual-information maximizer. A nuisance penalty compares the nominal and marginalized information, with the signed change also retained. Region-dependent variances can themselves convey information, so arbitrary heteroscedastic nuisance changes do not imply monotonic information loss.

D. Historical Frozen Paired Synthetic Evaluation

The earlier protocol reserved seeds 41000–41019 before evaluation, distinct from pilot seeds 31000–31005. Each method receives exactly eight acquisitions and 24 shared candidate opportunities per step, independent of truth and the method's posterior. Candidate and acquisition random streams are controlled separately. The then-executed inference and calibration controller remained active; the planning horizon is one. A second context changes acquisition velocity by $+2\%$, attenuation by $+4\%$ and noise by $\times2$, without giving these changes to the inference model. All 80 runs completed, with no execution or measurement-rejection failures. Source fingerprints were unchanged during evaluation. Table XII reports physical MAP error.

Define paired advantage as established error minus nuisance-aware error. The nominal mean advantage is 0.014 mm, with 95% paired-bootstrap interval $[-2.24, 2.94]$ mm: three wins, seven losses and ten ties. Under stress it is 33.58 mm, interval $[-30.62, 95.83]$ mm: ten wins and ten losses.

TABLE XII
HISTORICAL FIXED-CANDIDATE SYNTHETIC ABLATION. ERRORS IN MM; 20 SEEDS PER CONTEXT AND METHOD, EIGHT ACQUISITIONS PER RUN.

Context	Method	Mean	Median	P90	> 30 mm
Nominal	Established	15.12	13.85	23.80	1/20
Nominal	Nuisance	15.10	15.20	20.43	1/20
Stress	Established	180.14	191.93	285.29	17/20
Stress	Nuisance	146.56	126.36	275.54	17/20

Intervals use 10,000 paired-seed resamples. Both include zero; reliable superiority is not established. Both methods exceed 30 mm in 85% of stress cases. The new method reaches 426.96 mm error on seed 41011, worse than the baseline by 220.46 mm. No run crosses the predefined 0.70 false-confidence threshold, but stress abstention is only 1/20 versus 0/20. Low thresholded false confidence does not mean failures are recognized.

The supported outcome of this historical study is an auditable observation-level design calculation, not proven localization improvement or sim-to-real transfer. The method remains explicitly experimental. The complete artifact preserves per-step actions, errors, nuisance diagnostics, configuration and code hashes in `research_results/nuisance_design_validation.json`. A compact in-app report is generated from that artifact and hash-checked, rather than manually transcribing favourable results. Subsequent tuning requires development data and a new reserved confirmatory test set.

E. Scientific Interface

The redesigned interface represents the live posterior as a three-dimensional relief whose height is probability relative to the current peak, not physical defect depth. R90 contours are computed in millimetres from panel dimensions and discrete posterior cell centres. They are not field-calibrated coverage. Source/receiver markers reflect the recommended experiment; decorative moving excitation rings are explicitly illustrative. Global pause, reduced-motion support and off-screen animation suspension keep motion subordinate to the scientific readout. The interface separates the old unitless utility, the new model-information bits, posterior containment and actual benchmark error.

XVIII. CONCURRENT ACQUISITION AND EXTENDED REFERENCE STUDY

A. Implemented Time-Origin Contract

The later October v2 revision replaces the reference sketch's excite-then-record sequence. A classic ESP32 precomputes a quantized chirp or tone burst, records 16 pretrigger samples, and performs DAC25 source updates and ADC34 receiver sampling in one timed loop. GPIO26 marks excitation onset for external timing measurement. Rational sample deadlines avoid representing a truncated 62 μs interval as exactly 16 kHz. Frames retain sample call timestamps, ADC values, commanded DAC codes, the full experiment hash, fresh request identifier, trigger and drive intervals, ordered counts, integrity checksum and completion status. The host checks the complete declared excitation, timing deviations, waveform sequence and coverage before ingestion. The drive envelope is narrower than the simulated waveform bank: unsupported families are rejected before excitation.

The host requires a chip-specific operator declaration binding sampling rate, drive family and bounded clock/alignment errors to an external bench-report digest. A frame check is not a transfer-function measurement or authentication of the apparatus. The source has been compiled and tested against Arduino host mocks, including an independently generated mock frame accepted by the Python parser. No ESP32 board-target SDK compilation, flashing, external timing capture or physical inspection was performed. An isolated stable SDK installation attempt did not complete. Electrical DAC/ADC latency, analog filtering, amplifier limits, coupling and physical pose remain unmeasured.

The ingestion service, rather than client-supplied sensor names, assesses excitation-relative timing. Missing timing/adapter support gives zero structural evidence, blocks reference calibration and cannot satisfy verification, while preserving the recording for history and plots. A bounded exact external WAV declaration may instead identify a zero-trimmed trigger, source rate, complete experiment and external reference; this remains operator-declared evidence. Original checked serial timestamps, ADC values, DAC codes and declaration metadata are retained before ledger append. Regression tests verify that changes to retained timing evidence invalidate content verification.

B. Prediction-Oriented Reference Criterion

The optional fast selector evaluates every spatial grid cell under a nominated diagnostic source–receiver geometry. With ℓ_z its bistatic path, $g_z = (-\ell_z/v^2, 1)$ and speed/delay covariance C, define

$$W = \sum_z p_z g_z^T g_z, \qquad V = \operatorname{tr}(WC). \tag{52}$$

For each feasible direct reference, the implemented correlated Gaussian update predicts C_r and ranks $v^2 \operatorname{tr}(WC_r)$ plus configured acquisition cost. This is locally linear, posterior-weighted prediction-variance design, related to established V-optimal design; neither Gaussian conditioning nor prediction-variance minimization is asserted as newly invented. The objective is path-equivalent propagation uncertainty, not defect-position loss. The selector remains optional and distinct from the finite-horizon localization-risk controller.

C. Prospective Strong-Baseline Comparison

A source-frozen scalar study used 48 new independent seed clusters, four diagnostic geometries with nonuniform 20-by-20 spatial maps, two prespecified acquisition-noise regimes, five reference policies and three opportunities per policy: 1,920 attempted runs. Truth, innovations, candidate opportunities,

TABLE XIII
EXTENDED SCALAR TARGET ARRIVAL RMSE (μs)

Reference policy	Nominal noise	Doubled noise
Cyclic	229.82	419.26
Seeded random	232.30	396.96
Scaled A-optimal	200.23	363.06
Scaled D-optimal	190.85	342.63
Task-conditioned	189.09	343.47

initial covariance and geometry were sealed before any policy ran. Measurement noise uses fixed true acquisition physics and is never supplied to selection. Comparator A/D designs use dimensionless speed/delay covariance scales and the same declared cost conversion. All runs, costs and rejected updates remain in summaries; 1,909 runs completed without rejection and 11 opportunities were rejected.

Pooled comparator-minus-task RMSE advantages, with seed-cluster bootstrap 95% intervals, were 58.26 μs [35.13,82.52] for cyclic, 48.35 [26.69,70.91] for random, 15.37 [4.79,26.57] for scaled A-optimal, and 0.46 [-5.70,6.90] for scaled D-optimal. Intervals resample 48 clusters, preserve within-seed dependence and are exploratory without multiplicity correction. D-optimal differences remain inconclusive for both RMSE and prediction-plus-cost; task RMSE is slightly worse under doubled noise. A general superiority claim is unsupported.

Task nominal-noise timing-offset error worsens to 186.23 μs versus cyclic's 168.23, despite better target prediction. Its posterior-weighted 95% latent-arrival interval coverage is 91.83% nominal and 66.19% under doubled noise. These parameter intervals do not achieve nominal coverage and are not noisy-waveform or defect-confidence guarantees. All rejected updates and geometry/regime contrasts are retained in the full JSON. The study is synthetic scalar propagation evidence, not physical waveform or defect-localization validation. Earlier 192-run scalar and 48-run waveform studies retain their original source boundaries.

The v2 release passed 470 backend tests, with one deliberate legacy-pickle checkpoint skip, and 30 frontend tests, type checking, lint and production build. The complete study, protocol, source hashes and narrower claim assessment accompany the local review package. Close prior art already covers active Bayesian instrument calibration and acoustic reference localization; the specific apparatus/control interaction requires claim-level novelty and inventive-step review. Publication and grant probabilities cannot be inferred from software checks or these synthetic results.

XIX. MEASURED-SOURCE ASSESSMENTS: 8–9 OCTOBER 2026

Four separately scoped studies use published measured recordings to test screening, source localization, sequential path replay, and signal quality. Table XIV identifies their distinct evaluation units. Protocols fix grouping, preprocessing, models, and thresholds before the new fits; complete predictions, poor cases, and source/input hashes are retained in the four dated measured-study JSON reports. GFRP and Bologna are reanalyses of previously examined records, rather than independently acquired blind trials. The new OGW and MORPHO analyses use separate ultrasonic instruments. These studies have not exercised or validated the ARGUS physical apparatus.

TABLE XIV
MEASURED TASKS AND EVALUATION UNITS

Source	Physical groups / records	Task and reported metric
GFRP audio	8 plates / 2,568 taps	Point screening; balanced accuracy
Bologna	1 plate / 9 sources	Transient-source error, mm
OGW	1 plate / 4 mass positions; 18 banks	Offline path reveal; conditional MAP error, mm
MORPHO	5 panels / 1,011 selected labels	Operator quality; BAD recall

A. Whole-Plate Audio Screening

The TU Darmstadt GFRP record [23] supplies 2,568 unaugmented microphone taps from eight plates: 2,100 intact-labelled and 468 defect-labelled recordings. Each clip contains 5,120 samples at 51.2 kHz. The new study verifies publisher labels, original archive/member integrity, and decoded-recording duplicates, then recomputes the 15 fixed ARGUS features after the declared per-record resampling/filtering. Filenames, coordinates, plate identities, and labels do not enter the feature matrix. Four grouped folds keep complete plates apart; standardization uses training rows only. No previously fitted CORAL map, acoustic reference, or model checkpoint is used.

Fixed logistic regression yields pooled out-of-fold balanced accuracy 0.8381 and AUROC 0.9222, versus 0.5000 for the no-feature majority baseline. A 2,000-resample bootstrap of the eight complete plate clusters gives a balanced-accuracy interval [0.7521, 0.9154]. This interval conditions on the already fitted out-of-fold predictions, without model refitting. All 2,568 records receive predictions. Performance is heterogeneous: plate 5 has 188 false positives among 240 intact-labelled taps, whereas several other plates perform substantially better. The result supports dataset-conditioned audio screening, with eight specimen groups; it establishes neither a defect coordinate nor adaptive guided-wave inspection or field probability of detection.

B. Previously Seen Transient-Source Localization

The Bologna dataset [24] contributes nine measured source positions with three simultaneous channels per position on one aluminum plate. Its native rate is 2 MHz. Each fold holds out a complete position and fits timing or feature calibration on the other eight; channels from one event remain together. The fixed 5% cumulative-energy time picker feeds an isotropic time-difference-of-arrival model with training-fitted effective speed and relative sensor offsets.

Mean source error is 108.99 mm for calibrated TDoA, compared with 181.07 mm for the training-position centroid and 140.56 mm for three-nearest-neighbor regression. Only one of nine TDoA predictions lies within 50 mm. No independent repeat axis or second specimen is available. These descriptive improvements concern a transient source on a previously seen public record, not the location of a persistent hidden defect; no specimen-generalization interval is inferred from the nine positions.

C. Native Ultrasonic Offline Path Replay

The OGW temperature extension [31], [32] provides twelve transducers on a 500 by 500 mm CFRP plate. A protocol-index subset contains 18 complete banks at nominal 20, 40, and 60 °C, with 40 kHz excitation and native 10 MHz sampling. Each bank has 66 actually recorded unordered source–receiver pairs; reciprocal observations are not invented. Four reference locations use removable surface-mounted masses. Original ZIP-member lengths, CRC32, and local SHA-256 are verified; the whole-archive publisher MD5 remains unverified because only bounded members were downloaded.

Healthy thermal cycle 1 supplies reference waveforms; cycle 2 supplies reserved healthy tests in this dated replay. A position and all its temperatures are held out together. The scalar observation is $1-|\rho|$ between reference and catch waveforms. A fixed RAPID ellipse kernel, with $\beta = 1.05$, defines the spatial mean. Amplitude and residual noise are fitted only on the other three positions. Gaussian updates combine a healthy hypothesis with a uniform spatial prior; their conditional-independence assumptions and weights are not field-calibrated. Adaptive disagreement selects the next path from predicted response variance under current weights and observes only that selected test path. Cyclic, geometry-spread, training-static, and seeded random sequences share inference and reveal budgets. Sixteen random orders per case generate policy variability. The 960 trajectories reuse measured banks and are not 960 independent measurements or physical experiments.

At 32 revealed paths with condition-matched references, adaptive conditional MAP error is 160.76 mm versus 190.70 mm for random order; conditional 50 mm success is 25.00% versus 8.85%. These location scores are computed within the damage-cell map even when the separate damage decision is negative. Adaptive damage-positive rate is only 50%, while the healthy false-positive condition–model rate is 66.67%. The latter comprises twelve model evaluations reusing just three unique healthy test banks across four calibration folds. Thus, conditional localization improvement does not constitute successful operational NDE.

Full 66-path inference has mean error 213.28 mm and full-bank classical RAPID 209.43 mm under matched references. Adding paths does not repair the declared response model. Applying the 20 °C baseline at all temperatures gives adaptive error 200.04 mm and no 50 mm successes. Condition matching is not unseen-temperature generalization. One plate and four surrogate locations support descriptive model/policy comparison, not a large independent-sample efficacy claim or validation of the 16 kHz ARGUS probe.

D. Cross-Panel Operator Quality: Negative Transfer Result

The verified CC-BY-4.0 MORPHO curated record [33] contains 12,504 native 1 MHz traces from five panels, with 1,011 actively selected operator labels: 875 GOOD and 136 BAD. These labels concern clean, noisy, saturated, low-energy, or drifting waveforms. They are not defect positions or an unbiased acquisition sample. Sixteen waveform-only features from the first 500 samples feed fixed logistic regression and a 300-tree forest in five leave-panel-out folds. Training-only scaling, a majority dummy, and one within-training-panel shuffled-label forest provide controls. All labelled rows receive out-of-fold predictions.

At the frozen 0.5 threshold, the forest rejects only 1/136 BAD signals despite ordinary accuracy 0.8595 and AUROC 0.7715; balanced accuracy is 0.4997. Logistic regression rejects 27/136, with balanced accuracy 0.5124 and AUROC 0.2675. The majority dummy achieves accuracy 0.8655 by rejecting no BAD signals; the shuffled-training forest rejects none. FOD6 has only GOOD labels, so its fold AUROC and balanced accuracy remain undefined. Strong panel-dependent label prevalence and discrepancies between README cycle counts and actual HDF5 groups remain documented. No threshold was retuned on held-out labels, and no learned quality classifier was promoted to the live controller.

E. Healthy Reference Transport and Confirmation Gate

The false alarms motivate a separate, healthy-only development branch using established baseline stretch and phase-compensation principles [34], [35]. Recorded electrical pitch aligns source-clock delay before a bounded receiver time-scale, residual delay and gain are fitted to first-cycle healthy anchors at nominal 20, 40 and 60 °C. Prediction interpolates in actual mean surface temperature; it cannot fit a warp to a damaged/test catch. Immutable policy, template and source identities accompany each prediction. This remains an affine approximation to a dispersive, temperature-dependent wavefield.

The initial $\pm 1\%$ stretch domain qualified zero of 66 receiver paths; that failure and its exact source were archived. Expanding the healthy-development domain to $\pm 5\%$ with a fixed 41×41 search retained the same source-correlation, residual and identifiability limits. Mean healthy 40 °C leave-temperature-out DI decreased from 0.271 to 0.063. These diagnostics include *all* 66 paths, including unqualified fits; only 20 paths met qualification across the anchors. A 16-path horizon was therefore frozen from healthy evidence. The residual scale is a held-out healthy-temperature proxy, not an independently estimated noise distribution.

A new, explicitly exploratory reanalysis of the previously seen banks compares three likelihood variants on this common eligible path set and horizon. Adaptive conditional MAP errors are 159.45 mm for the raw global-scale model, 171.92 mm for transported references with global scale, and 170.03

mm with path-dependent residual scales. Thus, improved healthy waveform agreement did not improve localization in this development experiment. These are conditional-location diagnostics, not new blind tests or evidence that transport guarantees better decisions.

An experimental reference gate uses a complete healthy trajectory as one calibration unit. Each calibration score is the maximum, over the fixed budget prefixes, of the mean squared normalized selected-path DI. The current observed prefix score is compared with those complete-trajectory maxima. A rank comparison gives

$$p_{\mathrm{ref}} = \frac{1 + \#\{j : A_j \geq A_{\mathrm{current}}\}}{n+1}, \tag{53}$$

where each A_j belongs to an entire healthy block. Such ranking principles are established [36]; interpreting the complete-prefix maximum requires frozen control/reference identities and exchangeable healthy blocks. At least 19 blocks are required for 5% rank resolution. The initial development check reused only three old healthy banks and consequently returned unsupported gate decisions. Paths and Monte Carlo orders were not promoted into independent calibration blocks; that incomplete development result remains retained separately from the later confirmation.

The expanded verification suite passes 662 backend/acquisition tests, with one deliberate legacy-pickle skip, and 34 frontend tests, type checking, lint and build. These checks do not establish physical inspection efficacy.

The planned 29-bank confirmation completed on 9 October after temporary HTTP 403 download failures were resolved through a public retry. All 29 complete native banks, totaling 403,070,304 bytes, passed member-size/CRC and local SHA-256 verification. They comprise 19 healthy calibration blocks and ten separate 30/50 °C confirmation banks. The report filename retains the 8 October protocol tag. Predictions, fitting policy and controller parameters were frozen before reading confirmation catches; source hashes remained unchanged during scoring. The report retains 864 trajectories and 216 controller-bound calibrations, each using the same 19 physical blocks, with zero recorded rejections. These counts are not independent specimens or newly acquired adaptive experiments.

At 16 paths, adaptive conditional MAP errors are 161.20 mm for raw global scaling, 206.49 mm for transported global scaling, and 215.39 mm for path-dependent transported scaling. Transport therefore worsened this confirmation localization metric. Across all three adaptive variants, no allowed prefix produced a supported anomaly: the two unique healthy test banks had zero false alarms, but the eight mass-position/temperature conditions had zero detections. Zero false alarms in this setting reflects an insensitive, conservative gate and is not successful NDE. The new temperatures reuse the same plate and four removable-mass positions. Correlated cooling calibration blocks and unverified exchangeability still prevent a field-coverage claim.

For each transported random-order variant, 10 of 128 damaged-condition order runs produced an early supported alert, with none in 128 healthy-condition order runs. Averaging the 16 order repetitions within each of eight mass/temperature conditions first gives a 7.8125% mean any-prefix alert frequency. These are replays of the same measured banks, not ten independent physical detections. Raw random order and every geometry-spread variant produced no supported alerts. No variant retained a supported alert at the final budget; the reported any-prefix counts preserve early events that a final-only summary would miss.

Experimental dispositions request qualified references or diverse verification, with stale identity/index guards. They authorize neither an actuator nor a healthy structural release. The ordered clock/reference qualification and model/control feedback remain candidates for separate prior-art and human-inventorship review; BSS, phase compensation and conformal ranking are not claimed novel.

Measured-data replay consequently provides inspectable positive and negative development and confirmation evidence, while apparatus validation and operational NDE remain open.

XX. Conclusion

ARGUS NEO demonstrates an inspectable software loop for selecting the next acoustic experiment and checking the reliability of its evidence. It combines location and metrology blocks, finite diagnostic/reference selection, declared verification/abstention rules, and gated structural fusion. The optional controller evaluates reference calibration and diagnosis using at most two arrival-observation branches and a common physical localization loss. ARGUS-X preserves heuristic unsupported-state indicators, rejects duplicate or infeasible evidence, and provides a human-controlled emergency latch and replayable integrity record; it does not establish a healthy structural release. Historical matched-model results support faster modeled uncertainty concentration than simple policies, while localization-error superiority remains unproven. October controls improve specified software invariants without establishing inspection efficacy. A candidate practical use is orchestration for constrained or sparse sensing, contingent on measured benefit. The required next evidence is a pre-registered, specimen-held-out blind physical campaign with complete counterfactual coverage, independent truth, calibrated coverage, predefined losses, and component ablations. Technical novelty, human inventorship, ownership, and filing eligibility require separate professional review.

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APPENDIX A
DEFAULT CONFIGURATION

TABLE XV
IMPORTANT REPRODUCIBLE DEFAULTS

Subsystem	Parameter	Value
Panel	width × height	0.60×0.40 m
Material	velocity	180 m/s
Material	attenuation	1.65 m^{-1}
Material	resonance	3100 Hz
Material	damping	95 s^{-1}
Material	noise standard deviation	0.007
Material	system delay	0.8 ms
Signal	sample rate	16 kHz
Signal	record duration	0.12 s
Belief	grid; base temperature	20×20; 4.8
Stop	confidence; normalized entropy	0.72; 0.38
Stop	maximum experiments	12
Planner	candidate count; top hypotheses	48; 24
Planner	I, D, C, R weights	1, .35, .16, .22
Planner	coverage weight	0.22
NEO	risk; calibration; trust weights	.55; .65; .30
NEO	horizon; beam; rollout samples	1; 4; 6
NEO	calibration threshold; switch margin	.62; .05
OOD	caution; abstain thresholds	.55; .82
Safety	amplitude; frequency; duration	.75; 7 kHz; .35 s
Evidence	minimum evidence weight	.08
Easy	radii; severity; noise multiplier	.10–.15; .78–.95; .55
Medium	radii; severity; noise multiplier	.065–.11; .58–.82; 1.0
Hard	radii; severity; noise multiplier	.04–.075; .38–.64; 1.55
Reproducibility	default seed	7

APPENDIX B
INTERPRETATION GUIDE FOR DEMONSTRATIONS

An entropy decrease means probability mass has concentrated, not necessarily that the mode is correct. Confidence combines concentration and local mass and is OOD-capped, but it is not a certified probability of correctness. Ground truth must remain hidden until the controller stops. Reveal physical error using panel dimensions, not normalized distance. If entropy falls while revealed error remains high, diagnose mismatch or overconfidence rather than calling the run successful. Never describe a two-case matrix, one controlled mismatch case, or simulated calibration coverage as general performance.

A concise description for a judge is: "ARGUS is a research prototype that uses predicted responses of competing location hypotheses to select the next acoustic experiment, and can request calibration or withhold an unsupported decision. It records the action, measurement and posterior update for inspection." A defensible evidence statement is: "The dated release-readiness report records the completed software checks. Historical synthetic risk-controller results are mixed, and the controller remains optional. Physical accuracy and patentability remain unestablished."

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